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1. Data Collection

1.1 Research setting and development of the methodology

This study was conducted within the Value Chain Hackers Lab at Windesheim University of Applied Sciences. The lab functions as an applied research environment in which student teams work with real organisations on challenges related to transparency, sustainability, and system resilience. Research in this context is shaped by real-world constraints such as limited project durations, variation in research experience between students, stakeholder availability, and evolving institutional expectations. These conditions situate the study firmly in an uncontrolled setting and influenced both the design and execution of the research process.

The Graduation Project Plan defined the overall direction of the research. It established that the study would use qualitative methods supported by desk research and would include document analysis and interviews as core data sources. At the GP1 stage, the methodological direction was intentionally kept at a conceptual level. GP1 did not specify an interview instrument, nor did it define a final set of constructs or questions. This was a deliberate choice, as the theoretical foundations and constructs had not yet been sufficiently developed at that stage.

During GP2, the methodology was therefore developed through a structured, multi-phase process rather than implemented immediately. The first phase consisted of an extensive early-stage desk research process focused on identifying relevant theoretical perspectives and determining whether the research problem itself was legitimate and defensible. This process was not a simple literature search but an iterative exploration across multiple domains related to reproducibility, research transparency, documentation practices, research continuity, and applied research quality. The purpose of this phase was to test whether independent theoretical perspectives converged on similar problems and whether the challenges observed within the VCH Lab reflected broader structural issues rather than isolated student-level shortcomings.

This early-stage desk research involved deliberately exploring different conceptual lenses, comparing their explanatory value, and critically filtering out models that were not sufficiently relevant to the applied context of this study. Only after convergence emerged across multiple sources was a focused theoretical foundation established. This ensured that the constructs used in the study are not arbitrary but grounded in broader academic reasoning about how research processes fail, how knowledge is lost, and how methodological transparency can be strengthened.

The outcome of this process was then formalised through the development of an operationalisation table. In this table, the abstract constructs identified through desk research were translated into concrete dimensions, indicators, and observable elements. This step defined explicitly how concepts such as traceability, transparency, continuity, validation, and methodological consistency could be examined empirically. Importantly, the operationalisation did not merely define what should be examined, but also specified which data sources were appropriate for examining each construct and how those data should be interpreted.

A second desk research phase was subsequently conducted with a different focus. Whereas the first phase established conceptual grounding, the second phase focused on methodological legitimacy. This research examined academic studies that investigate documentation practices, reproducibility, research data management, and research process quality using interviews. Particular attention was given to studies that clearly described their interview protocols, topic structures, and reasoning behind instrument design. This phase served two purposes. First, it validated that using structured qualitative interviews is an appropriate method for investigating the type of practices examined in this study. Second, it provided concrete methodological examples that informed how the interview guide should be structured to achieve research-grade rigor rather than relying on intuitive questioning.

Only after completing both desk research phases and the operationalisation was the interview instrument designed. The interview guide was constructed by translating operationalised constructs into structured thematic blocks and specific questions. For example, constructs related to traceability were translated into questions about how decisions were documented and whether conclusions could be reconstructed. Constructs related to reusability were translated into questions about whether future teams could realistically continue the work based on existing documentation. Constructs related to system alignment were translated into questions about expectations, tooling, supervision practices, and handover processes. The structure of the guide reflects patterns identified in the methodological desk research, including the use of consistent core questions, probing questions, and lifecycle-oriented topics.

This sequencing demonstrates methodological development rather than deviation. The overall research design defined in GP1 was retained. What evolved during GP2 was the depth of conceptual grounding and the rigor with which the instruments were designed. The interview guide was not created in advance of the theory, but deliberately constructed only after constructs had been clarified and the appropriateness of the method had been validated. This strengthens the methodological quality of the study rather than weakening its alignment with the original plan.

1.2 Data sources, methods, and sampling in an uncontrolled setting

Three complementary sources of data were used in this study: desk research, document analysis, and structured interviews. Each source plays a distinct methodological role in relation to the research question, and together they form a coherent and integrated data collection strategy rather than separate activities.

Desk research formed the conceptual and methodological foundation of the study. The early-stage desk research established the legitimacy of the research problem and supported the identification of relevant theoretical constructs. The operationalisation then translated those constructs into dimensions and indicators that could be examined empirically. The subsequent methodological desk research validated the use of structured interviews and informed the design of the interview instrument. As a result, desk research did not function merely as background reading but as an active driver of research design, construct selection, and instrument development.

Document analysis was used to examine how research practices materialise in actual outputs produced within the VCH Lab. Five completed case studies were analysed, representing the Bakery, CRM, Organizational, Textile, and Cacao tracks. These cases were selected because they represent full project trajectories rather than partial deliverables, which makes them suitable for examining documentation practices, methodological consistency, continuity, and validation across an entire research process. Each case was analysed using the dimensions defined in the operationalisation table, including documentation completeness, traceability of reasoning, evidence of validation, reusability of outputs, and alignment between methods and conclusions. This ensured that each case was examined using the same theoretically grounded criteria rather than through subjective impression.

Structured interviews were conducted with the VCH Lab supervisor and four available former team leaders who had completed projects within the lab. Participants were selected purposively rather than randomly. The supervisor was included because of

their overview across multiple cohorts and projects and their ability to identify systemic patterns across cases. Former team leaders were included because of their direct experience with research execution, documentation practices, decision making, and handover. This sampling strategy prioritises relevance and informational richness over numerical representativeness, which is appropriate for qualitative research conducted in an applied and uncontrolled setting.

The interview guide used in the study was directly derived from the operationalisation and validated through the methodological desk research. Each thematic block in the guide corresponds to constructs identified through theory and desk research, such as documentation quality, traceability, continuity, reusability, and system alignment. The structure of the guide reflects research-grade interview practices identified in the methodological desk research, including consistent core questions, probing questions, and structured topic sequencing rather than open-ended conversational interviewing. This ensures that the data generated through the interviews is systematically aligned with the conceptual focus of the study and that the instrument itself meets academic expectations for rigor.

The same structured interview guide was used for all participants. This supports consistency across interviews and allows meaningful comparison between responses. The combination of desk research, operationalised document analysis, and theory-grounded structured interviews creates methodological coherence. Desk research defines what should be examined, document analysis shows how practices appear in real artefacts, and interviews provide insight into how those practices are understood, enacted, and constrained by those directly involved. Together, these data sources provide a coherent and sufficient empirical basis for answering the research question within the defined scope of the study.

1.3 Data quality, credibility, ethics, and bias

Ethical considerations were addressed throughout the data collection process. All interview participants were informed about the purpose of the study, the voluntary nature of their participation, and the academic use of their contributions. Verbal informed consent was obtained at the start of each interview. Participants were informed that their identities would not be disclosed and that excerpts would be anonymised in the report. For this reason, student names have been removed from all transcripts and are not referenced in this report. The document analysis relied on materials that were already part of the internal VCH Lab environment and that were accessed for the

purpose of educational evaluation and improvement. No sensitive personal data or confidential organisational information is disclosed.

Several concrete measures were taken to ensure that the data collected is usable, credible, and analytically meaningful rather than anecdotal.

First, methodological coherence was ensured through alignment between constructs, instruments, and data. The interview guide was not created intuitively but was constructed directly from the operationalisation table, which itself was grounded in the desk research. This means that each interview question corresponds to a construct that was theoretically justified and explicitly operationalised. This supports construct validity, because the study does not rely on vague opinions but collects data that is deliberately connected to what the research seeks to understand.

Second, consistency was built into the data collection process. The same structured interview guide was used for all five participants. This prevents the data from being shaped by ad hoc questioning or situational improvisation and instead ensures that differences in responses reflect differences in experience rather than differences in method. This functions as a reliability measure, because the data is generated under comparable conditions across participants.

Third, triangulation was deliberately embedded as a quality safeguard. The same underlying constructs are examined through three different types of material: desk research, document analysis, and interviews. For example, documentation quality is examined through case study artefacts and also explored through participant reflections on how documentation was handled in practice. Where patterns appear across sources, this strengthens the credibility of the findings. Where tensions appear between artefacts and accounts, these are treated as analytically meaningful rather than ignored. This use of cross source comparison functions as an ongoing validity check throughout the research process rather than as a post hoc justification.

Fourth, the credibility of the source material itself was considered explicitly. The five case studies analysed are complete project outputs that were formally archived in the lab repository and represent authentic research artefacts rather than draft or informal materials. The interview participants were selected because of their direct involvement in those projects. The desk research sources used for theoretical grounding and methodological justification were drawn from academic publications that explicitly address reproducibility, documentation practices, research process quality, and interview based methodology. These sources were not used for decoration but because they provided explicit methodological and conceptual guidance that shaped the design of the study.

Potential bias was also addressed explicitly. The researcher's close involvement with the VCH Lab creates a risk of confirmation bias and overfamiliarity. To mitigate this, personal impressions were not treated as data. Instead, claims in this study are grounded in concrete artefacts, recorded interview content, and structured analytical procedures. Where participant accounts contradicted what was visible in the documentation, these contradictions were not resolved in favour of prior assumptions but were preserved as empirical findings. This deliberate treatment of discrepancies as data rather than as inconvenience strengthens the integrity of the analysis.

1.4 Deviations, additional data collection, and adaptive methodological choices

The data collection process deviated in scope from the original plan outlined in GP1. The initial plan anticipated a larger number of interviews and the inclusion of a survey. In practice, data collection took place during the academic holiday period between late December and early January. This significantly limited the availability of potential participants. Rather than proceeding with a survey that would likely produce insufficient responses, the survey component was postponed. The interview sample therefore consisted of the VCH Lab supervisor and the four available former team leaders who had completed projects within the lab.

This deviation is explicitly acknowledged because it affects the breadth of the dataset. However, the reasons for the change are practical and acceptable within an applied research context. The sampling did not shift from purposive to convenience based. All participants were selected because they were directly involved in completed case studies and therefore had first hand experience with the practices under investigation. The supervisor contributed a system level perspective across cohorts. The team leaders contributed grounded perspectives based on direct execution of research work. This ensured that, despite the limited number of participants, the interviews generated information rich data that is closely aligned with the research question.

The reduced interview scope was deliberately compensated for through the integration of additional empirical and conceptual material. The document analysis of five completed case studies is treated as a substantive data source rather than as preparatory context. This analysis revealed recurring patterns across projects, including

loss of contextual information, undocumented methodological decisions, fragmented file structures, and weak evidence of validation. These patterns were identified before the interviews were conducted and directly informed the focus of the interview guide. This means that the interviews were not designed around abstract assumptions but were grounded in observed empirical issues.

In addition to the planned desk research used to define constructs and operationalisation, an additional desk research activity was conducted after the initial methodological design had been developed. This second desk research phase focused specifically on validating the appropriateness of the interview approach and the structure of the instrument. This step was taken because the operationalisation process revealed that the credibility of the instrument itself would be central to the credibility of the study as a whole. This constitutes additional data collection undertaken in response to emerging methodological needs rather than a fixed, pre scripted process.

This phase also required engagement with adjacent research domains, including research data management, lifecycle documentation practices, knowledge handover, and institutional continuity. These domains were not the original thematic focus of the project but proved directly relevant to understanding the mechanisms behind documentation breakdown and loss of research continuity. Their inclusion strengthened the conceptual grounding of the study and expanded the theoretical basis beyond a single disciplinary lens.

Taken together, these deviations and adaptive choices do not weaken the research design. They demonstrate responsiveness to practical constraints, methodological self critique, and deliberate strengthening of rigor. The final data collection process is therefore more theoretically grounded, more methodologically justified, and more closely aligned with the research problem than a mechanically executed original plan would have been.

2. Data Analysis

2.1 Analytical approach and reasoning logic

The analysis in this study was not conducted as open interpretation but followed a structured, theory grounded analytical logic. The purpose of the analysis was not to describe experiences in isolation but to identify patterns, mechanisms, and structural issues that explain why research outputs within the VCH Lab consistently lack transparency, traceability, and reusability.

The analytical process was guided directly by the constructs defined in the operationalisation. These constructs include dimensions such as documentation quality, traceability of decisions, continuity across project phases, methodological consistency, validation practices, and system level alignment. This means that the analysis did not begin with raw impressions but with a clearly defined conceptual lens derived from the desk research and operationalisation work.

All empirical material was analysed through the same conceptual structure. The five case studies were examined against the operationalised dimensions, and the interview data was interpreted using the same constructs. This ensured that analysis across sources remained coherent rather than fragmented. The aim was not to compare participants as individuals but to identify recurring patterns across artefacts and accounts.

The analytical reasoning followed a consistent logic. First, empirical material was examined for concrete evidence related to each construct. For example, when analysing traceability, attention was given to whether decisions, assumptions, and analytical steps could be reconstructed from documentation and participant descriptions. Second, evidence from different sources was compared. If a pattern appeared in both case study artefacts and interview accounts, it was treated as more robust. If discrepancies emerged between what participants reported and what was visible in documentation, these discrepancies were treated as analytically meaningful rather than ignored. Third, interpretations were only developed after evidence had been identified across multiple data points rather than based on isolated examples.

This approach ensures that claims in this report are grounded in data rather than opinion. The analysis therefore moves from concrete observations in artefacts and interviews, through structured interpretation using predefined constructs, toward conclusions that remain traceable to identifiable evidence.

2.2 Qualitative analysis of empirical material

The qualitative analysis focused on identifying recurring patterns across the document analysis and the interviews rather than treating each project or participant as an isolated case. This is consistent with the purpose of the study, which is not to evaluate individual student performance but to understand systemic characteristics of research practice within the VCH Lab.

Analysis of the five case studies revealed consistent structural weaknesses across projects. Documentation was frequently incomplete or fragmented, key methodological decisions were often not recorded, assumptions underlying conclusions were not

explicitly stated, and the reasoning chain from data to outcome was difficult to reconstruct. In multiple cases, it was not possible to determine why certain research choices had been made, which sources had been prioritised, or how conclusions had been validated. These observations were not based on subjective judgement but on systematic assessment against the operationalised dimensions applied to all five cases.

The interview data provided additional depth to these findings by revealing how these documentation practices emerged in practice. Participants consistently described uncertainty about what should be documented, inconsistency in expectations between supervisors and teams, and a lack of shared structure for storing and organising research materials. Several interviewees explicitly stated that documentation was often treated as a final reporting requirement rather than as an active part of the research process. Others indicated that time pressure and shifting project scopes led to pragmatic choices where documentation was deprioritised in favour of delivering visible outputs.

Importantly, the interviews did not simply confirm the document analysis but helped explain the mechanisms behind it. Where the case studies showed gaps in traceability, interviews revealed that students often did not experience a clear rationale for documenting methodological choices. Where artefacts showed fragmented file structures, interviews revealed that teams frequently developed their own ad hoc systems in the absence of shared guidance. This convergence between observable artefacts and participant explanations strengthens the credibility of the analysis because it demonstrates both structural evidence and experiential explanation.

The analysis also revealed that the problem is not primarily individual competence but structural conditions. Participants did not describe documentation breakdown as a result of negligence but as the result of unclear expectations, lack of shared standards, limited continuity between cohorts, and the absence of institutionalised research practices within the lab environment. This is a critical analytical shift because it reframes the problem from a student level deficit to a system level design issue.

At this stage, the analysis therefore moves beyond description and begins to demonstrate explanatory power. The patterns identified do not merely show that documentation is weak, but illuminate why it becomes weak under current structural conditions.

2.3 Patterns, insights, and perspective shifts

The analysis revealed that the most significant patterns are not isolated incidents but consistent structural mechanisms that shape how research is conducted within the VCH Lab. These patterns align closely with recognised phenomena described in the literature on research practice, documentation, and institutional structures (Chen et al., 2022; Lalu et al., 2023).

A first recurring pattern concerns unclear expectations around documentation practices. Across the interviews, participants consistently described uncertainty regarding what should be documented, when documentation should occur, and how detailed documentation was expected to be. This corresponds with broader findings that inconsistency in institutional guidance strongly shapes research behaviour (Chen et al., 2022). The absence of shared expectations therefore represents a systemic gap rather than an incidental issue. This pattern aligns with findings in Chen et al. (2022), who demonstrate that variability in student research practices is strongly influenced by unclear institutional guidance rather than individual capability. The present data reflects this dynamic in an applied university context, where participants reported that expectations shifted depending on supervisors and project contexts rather than being institutionally anchored.

A second structural pattern concerns documentation as a final output rather than an active research practice. Participants repeatedly described documentation as something that becomes relevant primarily during reporting phases rather than during ongoing decision making. This finding reflects the broader observation that research practices are often not embedded in routine behaviour when institutional reinforcement is weak (Lalu et al., 2023). The interview data therefore provides empirical confirmation that documentation is not currently embedded into everyday practice. This interpretation is further supported by Lalu et al. (2023), who show that barriers to rigorous research practices are structural and behavioural rather than individual failures. The participants in this study did not describe themselves as unwilling to document, but rather as operating within a system that does not structurally reinforce documentation as part of routine research behaviour.

A third pattern relates to breakdown of continuity between cohorts. Participants explicitly referred to loss of knowledge when teams changed, difficulty understanding prior work, and reliance on informal verbal explanations rather than structured artefacts. This directly corresponds to the continuity and handover dynamics described by Burnette et al. (2016), who demonstrate that role transitions and documentation continuity are legitimate and empirically studied dimensions of research quality. The presence of this pattern in both interviews and document analysis strengthens construct validity.

Crucially, these patterns produce a significant analytical shift. The evidence does not support an interpretation that individual students are failing to document appropriately. Instead, the data consistently supports a system level explanation: documentation breakdown emerges from unclear structures, inconsistent expectations, and lack of institutionalised practices. This shift from individual deficit to structural explanation is precisely what the desk research on problem legitimacy was designed to validate. Chen et al. (2022) and Lalu et al. (2023) both confirm that documentation gaps and practice inconsistency are recognised research phenomena rather than student specific shortcomings. This perspective shift is analytically important because it directly influences the nature of the solution that can be justified in later project phases. If the problem were individual competence, the solution would logically focus on training. The empirical evidence instead justifies a solution that focuses on system design, structural frameworks, and institutionalised research practices.

2.4 Gaps, limitations, and implications for further research

The analysis also requires explicit acknowledgement of the boundaries of the study. The realised interview sample was smaller than originally intended due to timing constraints, which limits the breadth of perspectives represented. This limitation affects the degree to which claims about frequency of experiences can be generalised. However, the analytical aim of this study is not statistical generalisation but structural understanding. The consistency of patterns across interviews and document analysis mitigates the impact of sample size on explanatory validity. This use of triangulation aligns with established qualitative research standards (Kallio et al., 2016; DeJonckheere and Vaughn, 2019).

A second limitation concerns the absence of the originally planned survey instrument. While the initial design anticipated the use of surveys to capture broader perceptions, the current dataset relies on depth rather than breadth. This limitation restricts the ability to quantify how widespread specific perceptions are across the broader student population. However, the depth of qualitative insight remains strong and aligns with the methodological legitimacy of semi structured interviews as defended by Kallio et al. (2016) and DeJonckheere and Vaughn (2019), both of whom emphasise that rigor in qualitative research comes from coherence between research questions, instrument design, and analytical depth rather than from sample size alone.

The analysis also identifies implications for further research within the same project. Several patterns, such as differences between supervisors in expectations, or variations between tracks in documentation maturity, emerged but could not be fully explored

within the current scope. These observations do not weaken the current findings but instead indicate where additional data collection could meaningfully deepen understanding in later phases of the graduation project.

Importantly, the limitations identified do not undermine the central analytical conclusions. The evidence remains coherent across data sources, grounded in validated constructs, and aligned with recognised research phenomena. The study therefore remains methodologically defensible while being transparent about its scope.

3. Conclusions and Recommendations

3.1 Answers to the research question

The central research question guiding this study is:

How can the Value Chain Hackers Lab design and validate a reproducible research framework that makes student projects transparent, verifiable, and reusable in a consistent and institutionally reliable way?

The findings provide a clear and evidence grounded answer. The research demonstrates that the current limitations in transparency, traceability, and reusability are not caused by individual student shortcomings but by structural weaknesses in how research practices are organised within the lab. These weaknesses include unclear expectations regarding documentation, lack of shared methodological structures, absence of consistent handover practices, and insufficient embedding of documentation into daily research activity. These patterns were consistently identified across document analysis and interviews.

This means that a reproducible research framework must not function as an optional tool or guideline but as a structural system that shapes how research is conducted throughout the entire project lifecycle. The evidence supports that such a framework must provide explicit guidance on what must be documented, when it must be documented, how it must be stored, and how continuity between cohorts must be safeguarded. Without structural integration, documentation practices remain dependent on individual interpretation and time pressure, which leads to the recurring breakdowns identified in the data.

The research therefore supports the conclusion that the solution is not educational in nature but design oriented. The lab requires a system that embeds reproducible

research practices into its operational structure rather than relying on individual motivation or ad hoc supervision.

3.2 Synthesis and strength of conclusions

The strength of the conclusions rests on the convergence of evidence across multiple sources and levels. Document analysis demonstrated that research artefacts frequently lack traceable reasoning, coherent structure, and reusable documentation. Interviews confirmed that this is not due to negligence but due to uncertainty, inconsistent expectations, and lack of shared systems. Desk research validated that these phenomena are recognised structural problems in research practice rather than context specific anomalies (Chen et al., 2022; Lalu et al., 2023).

This convergence creates a coherent explanatory model. Structural conditions shape behaviour. When documentation is not embedded in the system, students treat it as peripheral. When expectations differ across supervisors, teams develop inconsistent practices. When handover is not structurally designed, knowledge loss becomes inevitable. These mechanisms explain the patterns observed in both artefacts and participant accounts.

The conclusions are therefore not speculative. They are grounded in three independent layers of evidence: theory, artefacts, and participant experience. This triangulation strengthens the internal validity of the study and supports the legitimacy of moving from analysis to design requirements.

At the same time, the conclusions remain bounded to the research context. The findings do not claim universal applicability to all research labs. They are specific to the operational conditions of the VCH Lab. However, because the mechanisms identified align with phenomena documented in academic literature, the conclusions extend beyond anecdote and can be treated as credible within the applied research context.

3.3 Specifications and criteria for solution development

The conclusions lead directly to concrete, research grounded specifications for the development of a reproducible research framework.

First, the framework must provide explicit documentation standards. The evidence shows that students currently do not share a clear understanding of what constitutes adequate documentation. The framework must therefore define required elements such

as documentation of assumptions, methodological decisions, data provenance, analytical reasoning, and changes in direction over time. These requirements must be expressed not as abstract principles but as concrete practices that can be followed and assessed.

Second, the framework must establish structural continuity mechanisms. The data demonstrates that loss of knowledge between cohorts is a persistent problem. The framework must therefore include requirements for project onboarding, handover documentation, shared repositories, and continuity of research artefacts across semesters. Without this, projects will continue to reset rather than accumulate knowledge. Third, the framework must be embedded into the research workflow rather than positioned as an add on. Interviews consistently showed that documentation is currently treated as a reporting task rather than as part of research practice. The framework must therefore be designed to integrate with how students already work, including templates, structured workspaces, and checkpoints that make documentation unavoidable rather than optional.

Fourth, the framework must support consistent supervisory alignment. Variation between expectations of different supervisors was identified as a contributing factor to inconsistency. The framework must therefore function as a shared reference point for both students and supervisors, creating a common standard that reduces interpretive variation. These specifications are not arbitrary design preferences. They are derived directly from the patterns identified in the analysis and therefore represent criteria that are empirically justified rather than conceptually idealised.

3.4 Validity, certainty, and remaining unknowns

The conclusions of this study are valid within the defined scope of the research. The patterns identified are consistent across data sources and are grounded in both empirical evidence and existing literature. This supports a high level of internal coherence and explanatory validity.

However, the study does not eliminate all uncertainty. Several questions remain open and are explicitly acknowledged. The extent to which different tracks within the lab vary in their documentation maturity was observed but not fully explored. The influence of individual supervisors on practice consistency emerged as a potential factor but requires further investigation. The long term effectiveness of any proposed framework cannot be established within this study alone and will require evaluation during the implementation phase of the graduation project.

These uncertainties do not weaken the conclusions but instead clarify their boundaries. The research provides a strong, evidence grounded foundation for framework design, while also identifying where further validation and iteration will be necessary. This transparency strengthens the credibility of the work rather than undermining it, because it avoids overstating what the data can support.